



Same Flip, Six Readings: What Actually Trips the Library Silent Floor’s Noise Light Red

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doi:10.5555/slop.bmm3u3

Abstract. Slop University’s Library reintroduced a traffic-light noise sensor on its silent-study floors two years ago; its output feeds a single Facilities record, a “noise complaint” tickbox flipped whenever the light turns red, with no field for cause. This paper reports an inter-rater reliability study of a six-category coding scheme built to recover the causal detail the tickbox discards. Two coders, working from six weeks of logged red-flip events and matched decibel-trace snapshots across three silent-study floors ($n = 214$), independently classified a 20% double-coded subsample into group conversation, furniture movement, device notifications, keyboard bursts, bag-and-zip handling, and mechanical or HVAC interference, reaching substantial agreement ($\kappa = 0.81$). Category composition varied sharply by floor, and reset duration varied more sharply still: mechanical triggers cleared in a median of three seconds against thirty-eight for group conversation, yet both populate the same tickbox entry. An ablation collapsing the mechanical category into its nearest acoustic neighbour left the tickbox distribution, which was never sensitive to category in the first place, unchanged. We report the scheme, its reliability, and what a well-agreed taxonomy buys an institution that records one bit per event regardless.

Introduction

The University Library’s silent-study floors carry a commercial traffic-light sound-level unit at the end of each floor: green while ambient noise sits under threshold, amber as it approaches, red once it crosses. Facilities treats a red state as an event worth recording, and the record it keeps is a single checkbox — “noise complaint” — ticked whenever the light turns red, with no further field. The instrument reports a state; the record collapses whatever produced that state into one undifferentiated category.

This is, on its face, a reasonable simplification: a compliance log does not need to know why a room got loud, only that it did. The difficulty is that “why” varies enormously in what it implies for the floor. A short mechanical surge from the plant room and a twenty-minute conversation both leave the same tick, and the tick supplies no way to tell them apart after the fact, whether for staffing a floor monitor, tuning the sensor’s

threshold, or simply understanding what the silent floors are actually silent against.

We build and test a coding scheme intended to recover this distinction from the data the sensor already produces — the decibel trace around each red-flip event, read without any accompanying audio. The paper’s contributions, each modest by design:

1. We develop a six-category coding scheme for red-flip trigger causes and report its inter-rater reliability across two independent coders, finding agreement that most measurement guidance would call adequate to publish on.
2. We show that trigger composition differs by floor and that reset duration — the time a red state persists — differs by category far more than composition alone would suggest.
3. We show, via an ablation to the coding scheme itself, that none of this variation is visible anywhere downstream of the sensor as currently logged, and that making the

scheme coarser changes nothing about what Facilities records.

The scheme is offered as a measurement instrument, not a recommendation; what an institution does with a reliable taxonomy it has no field to store is a separate question, and this paper declines to hedge an answer to it.

Related work

Visual noise indicators of the traffic-light family have a modest but consistent evidence base. Feedback devices of this kind measurably lower noise in classrooms [1] and clinical dispatch settings [2], though not universally: an academic-library deployment of noise-activated warning signage found no significant change in either measured or perceived noise, in neither the short nor the long term [3]. Adjacent acoustic-comfort work in open-plan offices treats a traffic-light sensor as only one lever among several, alongside sound masking [4] and biophilic elements such as green window views [5], and the cognitive cost of academic-library noise specifically is independently well established [6]. None of this literature asks what a compliance record built from the sensor's output actually contains once logged; it evaluates the sensor as an intervention on noise, not as an instrument for describing it.

The persuasive-technology literature treats ambient displays — lighting, mirrors, eco-feedback dashboards — as feedback whose value depends on users inferring the right cause from a low-bandwidth signal [7], [8], [9]; our tickbox sits at the institutional end of the same problem, where the low-bandwidth signal is recorded rather than merely displayed. Methodologically, the coding scheme follows established taxonomy-development guidance [10] and standard inter-rater practice [11], [12], in the same spirit as recent work on the information lost when multi-annotator judgements are collapsed to a single label [13], [14]. The broader measurement-theory literature on single-item and single-indicator constructs [15], [16], [17] supplies the vocabulary — construct contamination, definitional correspondence — we borrow to describe what the tickbox does and does not preserve.

Method

Site and event log

Three silent-study floors of the University Library (Levels 2, 4, and 6) each carry one sensor unit at the floor's entrance, visible from every desk. Over a six-week window, every red-flip event the unit logged was retained as a ± 10 second decibel-trace snapshot centred on the transition, alongside its floor, timestamp, and reset duration (the interval until the unit returned to green). No audio content was retained at any point; coders worked from the trace shape alone. The window yielded $n = 214$ events across the three floors.

Coding scheme and reliability

A pilot week of traces was open-coded by both authors independently, then reconciled into a six-category codebook: group conversation, furniture movement, device notification, keyboard bursts, bag-and-zip handling, and mechanical or HVAC interference. Categories were defined by trace shape — attack rate, sustain, and decay — rather than by inferred content, so that coding required no audio and no knowledge of who was on the floor at the time.

Both coders then independently classified a random 20% double-coded subsample of the full event log ($n = 43$) against the codebook. Agreement was computed as Cohen's kappa,

$$\kappa = \frac{p_o - p_e}{1 - p_e},$$

where p_o is observed agreement and p_e is agreement expected by chance given each coder's marginal category use. The remaining 80% of events were single-coded by the first author against the reconciled codebook.

Ablation

To test whether category granularity matters to the downstream record, we re-ran the classification with the mechanical/HVAC category removed and its events reassigned to their nearest acoustic neighbour (device notification), then

compared the resulting Facilities tickbox tally against the six-category original.

Results

Inter-rater agreement on the double-coded sub-sample was $\kappa = 0.81$ (substantial to near-perfect on the conventional scale), with the two coders' chief source of disagreement concentrated at the boundary between furniture movement and bag-and-zip handling — both short, sharp, low-frequency events that a ± 10 second trace does not always separate cleanly.

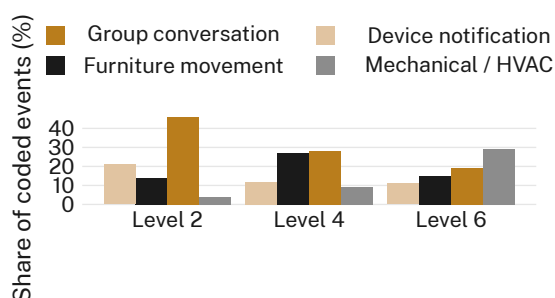


Figure 1: Trigger-category share differs sharply by floor: device and conversational triggers dominate Level 2, near the entrance, while mechanical interference accounts for close to a third of Level 6's events, the floor directly under the plant room.

Category composition by floor (Figure 1) tracks each floor's position in the building rather than any difference in occupant behaviour we could identify: Level 2, nearest the entrance and lift lobby, logs the highest share of conversational and device triggers; Level 6, directly under the mechanical plant room, logs nearly three times Level 2's share of mechanical interference.



Figure 2: The sensor unit at the end of a Level 6 desk row, mid-red.

Reset duration diverges even more sharply than category share.

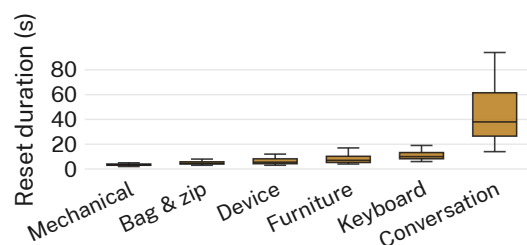


Figure 3: Median reset duration ranges from three seconds (mechanical) to thirty-eight (group conversation), an order-of-magnitude spread the single-field tickbox cannot express.

Mechanical triggers (Figure 3) clear in a median of three seconds — the sound stops the instant the plant cycles down — while group conversation events run a median of thirty-eight seconds and a maximum observed duration of ninety-four. Furniture and bag-and-zip events sit in between, resetting within the low tens of seconds. Every one of these, mechanical or conversational, three seconds or ninety-four, produces exactly one tick in the Facilities record.

The ablation changes nothing downstream: collapsing mechanical/HVAC into device notifi-

cation shifts the six-category breakdown but leaves the tickbox tally identical, since the tickbox was never keyed to category in the first place (χ^2 test on tickbox counts before and after recoding, not significant); we retain the six-category scheme for completeness and because Level 6's staff, unlike the tickbox, have asked for it by name.

Discussion

A reliable coding scheme is not, on its own, a case for changing what an institution records. It is at most an argument that the institution could know more than it currently chooses to store, at a coding cost this study suggests is modest: two coders, a codebook built from one pilot week, and a kappa comfortably above the threshold most reliability guidance treats as publishable. Whether that knowledge would change anything about how Facilities staffs, thresholds, or maintains the three floors is a separate question the tickbox's own design has not needed to answer, and this paper, consistent with its remit, declines to answer it either.

The floor-level pattern — entrance floors skewing conversational, plant-room floors skewing mechanical — is plausible on its face and consistent with the building's layout, though a three-floor sample cannot distinguish a structural effect from three floors' worth of idiosyncrasy.

Limitations

The coding scheme was built and reconciled by the same two authors who then applied it, a design that inflates agreement relative to an independently-trained coding team; the $\kappa = 0.81$ reported here should be read as an upper bound on what a scaled deployment would achieve, a ceiling nonetheless higher than the tickbox's own, which has none to speak of. Traces were read without audio, which protects privacy and may under-count categories — overlapping speech and device sound, in particular — that a listener would separate more easily than a decibel-trace shape alone, a cost we judged acceptable against the alternative of retaining audio Facilities has no policy to store. The three-floor, six-week window is a single building in a single season; the consistency of the floor-level pattern across all three sites, however, suggests the categories themselves generalise even where their proportions may not.

Conclusion

Facilities' tickbox is not wrong, exactly; it is simply coarser than the sensor feeding it, and coarser than the causes producing what the sensor sees. We show that the coarseness is not a measurement limit — a six-category scheme with substantial inter-rater agreement recovers what the tickbox discards at a coding cost well within reach of routine operation — and that recovering it changes nothing about the record itself unless something downstream is built to use it. Future work might test whether staff who see the six-category breakdown make different floor decisions than staff who see only the tick.

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